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INFUSION PUMP SYSTEM AND METHOD

Abstract

Some embodiments of an infusion pump system may be configured to wirelessly communicate with other devices using near field communication (NFC). In particular embodiments, by incorporating near field communication capabilities into the infusion pump system, user communications with the infusion pump system can be enhanced and simplified.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/988,081, filed Dec. 19, 2024, which is a continuation of U.S. patent application Ser. No. 18/667,929, filed May 17, 2024, which is a continuation of U.S. patent application Ser. No. 17/445,819, filed Aug. 24, 2021, now U.S. Pat. No. 12,064,591, issued Aug. 20, 2024, which is a continuation of U.S. patent application Ser. No. 15/893,145, filed Feb. 9, 2018, now U.S. Pat. No. 11,147,914, issued Oct. 19, 2021, which is a divisional of Ser. No. 15/383,176, filed on Dec. 19, 2016, now U.S. Pat. No. 10,207,047, issued on Feb. 19, 2019, which is a continuation application of and claims priority to U.S. application Ser. No. 13/946,330, filed on Jul. 19, 2013, now U.S. Pat. No. 9,561,324, issued on Feb. 7, 2017, the disclosure of each of which is hereby incorporated herein in its entirety by this reference.

TECHNICAL FIELD

[0002] This document relates to an infusion pump system, such as a portable infusion pump system for dispensing a medicine.

BACKGROUND

[0003] Pump devices are commonly used to deliver one or more fluids to a targeted individual. For example, a medical infusion pump device may be used to deliver a medicine to a patient as part of a medical treatment. The medicine that is delivered by the infusion pump device can depend on the condition of the patient and the desired treatment plan. For example, infusion pump devices have been used to deliver insulin to the vasculature of diabetes patients so as to regulate blood-glucose levels.

[0004] Users of infusion pump devices often need to communicate with the infusion pump via a user interface to control the operations of the infusion pump in a safe and effective manner. For example, a user may press a series of buttons on the user interface to enter food intake data into the infusion pump, such as a number of grams of carbohydrates that is indicative of a recently or soon-to-be consumed meal. The food intake data can be combined by the infusion pump system with other parameters to calculate a suggested bolus dosage of insulin based on the grams of carbohydrates entered by the user. In another example, a user may enter information into the infusion pump system via a user interface that indicates that the user is going to perform a level of physical exercise. In some circumstances, the infusion pump system may reduce the amount of a planned dispensation of insulin in response to the exercise information entered by the user.

SUMMARY

[0005] Some embodiments of an infusion pump system may be configured to send and receive data communications using near field communication (“NFC”) technology. By incorporating NFC technology within the infusion pump system, user communications with the pump system can be enhanced and simplified. For example, NFC can facilitate the convenient sharing of user commands or other data to an infusion pump system a NFC that is equipped with NFC

functionality. In some embodiments, pre-programmed NFC communicator devices (“NFC tags”) can be used to transfer data from the NFC tag to the infusion pump system via a simple hand motion or the like by the user of the infusion pump system. The data that is transferred may cause the infusion pump system to execute particular operations as defined by the data or in correspondence to the data. For example, a NFC tag can be configured to communicate a set of user input commands to an infusion pump system (e.g., user input commands that might otherwise be input via a series of menu selections and data entry steps on the user interface of the pump system) so as to rapidly indicate to the pump system that particular amount of food or carbohydrates will be consumed for a meal. In some embodiments, data can be written from the infusion pump system to a NFC tag. For example, a back-up copy of user settings that are used to configure an infusion pump system for a particular user may be downloaded using NFC from the infusion pump system and saved onto a NFC tag. In particular embodiments, the infusion pump system can be equipped with one or more accelerometers that can be used to activate the potential for NFC communications to take place when an acceleration at or above the threshold level is detected.

[0006] In particular embodiments, a medical infusion pump system may include a portable pump housing that defines a space to receive a medicine. The infusion pump system may include a pump drive system to dispense medicine from the portable housing when the medicine is received in the space. The infusion pump system may further include control circuitry that communicates control signals to the pump drive system to control dispensation of the medicine from the portable housing when the medicine is received in the space. Optionally, the infusion pump system may also include a near field communication (NFC) circuit electrically connected with the control circuitry to communicate infusion pump task data to the control circuitry. The NFC circuit can be configured to wirelessly receive the infusion pump task data from a NFC communicator device when the NFC circuit and NFC communicator device are positioned within a NFC proximity range.

[0007] In some implementations, the system may optionally include the NFC communicator device that is separate from the pump housing. For example, the NFC communicator device can be a near field communication tag storing the infusion pump task data. The infusion pump task data may comprise a unique identifier that identifies the near field communication tag, and in response to receiving the unique identifier, the control circuitry may execute user interface operations that correspond to the unique identifier. Optionally, the user interface operations may comprise user interface settings for calculating a suggested bolus dispensation of the medicine. In another option, the infusion pump system may further include an accelerometer electrically connected to the control circuitry, wherein the accelerometer may be configured to detect acceleration movement of the portable housing and to communicate the detected movement to the control circuitry. In a further option, the control circuitry is configured to compare a characteristic value of the detected movement to a threshold movement value. The control circuitry may be configured to activate near field communication with the NFC communicator device based on the comparison of the characteristic value to the threshold movement value. Optionally, the control circuitry may be housed in a controller housing that is removably attachable to the portable housing.

[0008] In some implementations described herein, the system may optionally include a remote-control device that is separate from the pump housing. The remote-control device can be configured to wirelessly communicate with a wireless communication device connected to the control circuitry (for example, a wireless communication device that is different from the aforementioned NFC circuit). Optionally, the remote-control device may further include a second NFC circuit that is configured to wirelessly receive the infusion pump task data from the NFC communicator device when the second NFC circuit and NFC communicator device are positioned within the NFC proximity range. In some cases, the NFC proximity range has a maximum working distance of less than 12 inches. The infusion pump task data may be indicative of a value of carbohydrates of a food item.

[0009] In particular embodiments, a medical infusion pump system may include a pump device and a controller device. The pump device may include a pump housing that defines a space to receive a medicine, and a drive system positioned in the pump housing to dispense the medicine from the pump device when the medicine is received in the space of the pump housing. Optionally, the controller device may be removably attachable to the pump device. For example, the controller device may be removably attachable to the pump housing so as to electrically connect with the pump device. The controller device may house control circuitry configured to communicate control signals to the drive system positioned in the pump housing to control dispensation of the medicine from the pump device. The controller device may also house a NFC circuit electrically connected with the control circuitry to communicate infusion pump task data to the control circuitry. Optionally, the NFC circuit is configured to wirelessly receive the infusion pump task data from a NFC communicator device when the NFC circuit and NFC communicator device are positioned within a NFC proximity range.

[0010] In some implementations, the system may further comprise the NFC communicator device that is separate from the pump device and the controller device. For example, the NFC communicator device may be a near field communication tag storing the infusion pump task data. Optionally, the infusion pump task data may comprise a unique identifier that identifies the near field communication tag, and in response to receiving the unique identifier, the control circuitry may execute user interface operations that correspond to the unique identifier. In one example, the user interface operations may comprise user interface settings for calculating a bolus dispensation of the medicine. Optionally, the system may further include at least one accelerometer electrically connected to the control circuitry. The accelerometer may be configured to detect acceleration movement of the portable pump housing and to communicate the detected movement to the control circuitry. The control circuitry may be configured to compare a characteristic value of the detected movement to a threshold movement value. The control circuitry may be configured to activate near field communication with the NFC communicator device based on the comparison of the characteristic value to the threshold movement value.

[0011] In various implementations of the system, the pump device may optionally be a one-time-use device equipped with one or more structures configured to prevent reuse of the pump device. Also, in some implementations, the controller device may optionally be a reusable controller device. For example, the controller device may include one or more of: a controller housing that is removably attachable to the pump housing in a fixed relationship; one or more electrical contacts disposed on the controller housing, the electrical contacts of the controller device being engageable with corresponding electrical contacts of the pump device when removably attached.

[0012] Additionally, particular embodiments described herein may include a method of controlling a portable infusion pump system. The method may include receiving input via near field communication (NFC) from a NFC tag storing data indicative of a task associated with using the portable infusion pump system. The method may optionally include controlling the portable infusion pump system to change an operation of the portable infusion pump system in based upon the data the input from the NFC tag. In some implementations, the method may further comprise prompting a user via a user interface display to confirm the operation change of the portable infusion pump system in response to receiving the input from the NFC tag. For example, the operation change to be confirmed via the user interface may include calculating or initiating a bolus dispensation of a medicine from the portable infusion pump system.

[0013] Some or all of the embodiments described herein may provide one or more of the following advantages. First, some embodiments of the infusion pump system may be configured to send and receive data communications using NFC technology. Second, some embodiments of an infusion pump system equipped with NFC technology may facilitate convenient user input of information to the infusion pump system. Third, the safety and efficacy of an infusion pump system may be enhanced because the rapid manner of inputting data to the infusion pump using NFC may

facilitate more timely and complete data entry by the user. Fourth, in some circumstances, some users who may not be mentally or physically able to reliably operate a conventional user interface of an infusion pump system may be able to reliably input data to an infusion pump system using NFC communication interface. Fifth, the infusion pump system equipped with NFC equipment may be configured to be portable, wearable, and (in some circumstances) concealable. For example, a user can conveniently wear the infusion pump system on the user's skin under clothing or can carry the pump system in the user's pocket (or other portable location) while receiving the medicine dispensed from the pump device.

[0014] It should be understood from the description herein that the term “NFC” (as used herein) or “NFC” capability (as used herein) is different from traditional radio frequency identification (“RFID”). For example, NFC is a more specific version of wireless communication that can be configured for one-way or two-way communications and that operates at a maximum range of less than about 12 inches, about 8 inches or less, and preferably about 4 inches or less (e.g., unlike the much greater communication range of the traditional RFID technology that extends for many feet or more).

[0015] The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a perspective view of an infusion pump system with NFC capabilities in accordance with some embodiments.

[0017] FIG. 2 depicts an assortment of example NFC tags that can be used with the infusion pump systems described herein.

[0018] FIG. 3 is a flowchart describing a process of using an infusion pump system equipped with NFC capabilities in accordance with some embodiments.

[0019] FIG. 4 is a schematic diagram of an infusion pump system with NFC capabilities in accordance with some embodiments.

[0020] FIG. 5 is an exploded perspective view of another infusion pump system with NFC capabilities in accordance with some embodiments.

[0021] FIG. 6 is a perspective view of another infusion pump system with NFC capabilities in accordance with some embodiments.

[0022] FIG. 7 is a perspective view of another infusion pump system with NFC capabilities in accordance with some embodiments.

[0023] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0024] Referring to FIG. 1, an infusion pump system **10** can include a portable pump **60** used to supply insulin or another medication to a user via, for example, an infusion set **70** (also referred to herein as infusion pump system **70**). In some embodiments, the portable pump **60** includes a user interface **62** comprised of input devices such as buttons **63a**, **63b**, **64a**, **64b**, **64c** and output devices such as display **65**. At least a portion of the user interface **62** is coupled to a pump housing structure **66** of the portable pump **60**, which houses the control circuitry for the portable pump **60**. In particular embodiments, the portable pump **60** may further include a NFC circuit **40** that facilitates short-range wireless communications between the internal control circuitry of the portable pump **60** and an external device such as a NFC tag **45**.

[0025] NFC can be used, for example, to rapidly input user commands or other data into the portable pump **60**, thereby at least partially reducing the need to actuate the buttons **63a-63b**, **64a-**

64c or other components of the user interface **62**. As explained further herein, the data input to the portable pump **60** via NFC may cause the portable pump **60** to execute particular actions, such as automatically calculating an amount of a recommended bolus delivery of insulin (or another medication) and prompting the user with an option to confirm and initiate such a bolus delivery. By incorporating NFC equipment within the infusion pump system **10**, user communications with the portable pump **60** can be enhanced and simplified. As a result, the accuracy and completeness of the data entered by the user into the portable pump **60** can be improved, and the user can experience greater convenience and time efficiency. Optionally, the portable pump **60** can further include an accelerometer **50** arranged in the pump housing structure **66**. In some embodiments, the accelerometer **50** can be used to activate the NFC communications when an acceleration at or above the threshold level is detected, as explained further below.

[0026] The infusion pump system **10** is configured to controllably dispense a medicine to be infused into the tissue or vasculature of a targeted individual, such as a human or animal patient. In some embodiments, the portable pump **60** includes the pump housing structure **66** that defines a cavity in which a fluid cartridge (not shown) can be received. For example, the fluid cartridge can be a carpule that is either user-fillable or is preloaded with insulin or another medicine for use in the treatment of Diabetes (e.g., BYETTA®, SYMLIN®, or others). Such a cartridge may be supplied, for example, by Eli Lilly and Co. of Indianapolis, IN. Other examples of medicines that can be contained in the fluid cartridge include: pain relief drugs, hormone therapy, blood pressure treatments, anti-emetics, osteoporosis treatments, or other injectable medicines. The fluid cartridge may have other configurations. For example, in some embodiments the fluid cartridge may comprise a reservoir that is integral with the pump housing structure **66** (e.g., the fluid cartridge can be defined by one or more walls of the pump housing structure **66** that surround a plunger to define a reservoir in which the medicine is injected or otherwise received).

[0027] The portable pump **60** includes a cap device **68** to retain the fluid cartridge in the cavity of the pump housing structure **66** and, optionally, to penetrate a septum of the fluid cartridge for purposes of establishing fluid communication with the infusion set **70**. The portable pump **60** includes a drive system (one example is described in more detail below in connection with FIG. 5) that advances a plunger in the fluid cartridge so as to dispense fluid therefrom. In some embodiments, the dispensed fluid exits the fluid cartridge, passes through a flexible tube **72** of the infusion set **70** to a cannula housing **74** retained to the user's skin **20** by a skin adhesive patch **78**. The dispensed fluid can enter through the skin **20** via a cannula **76** attached to the underside of the cannula housing **74**.

[0028] In some embodiments, the infusion pump system **10** can be configured to supply scheduled basal dosages of insulin (or another medication) along with user-selected bolus dosages. The basal delivery rate can be selected to maintain a user's blood glucose level in a targeted range during normal activity when the user is not consuming food items. The user-selected bolus deliveries may provide substantially larger amounts of insulin in particular circumstances in which the user's blood glucose level requires a significant correction. In some embodiments, the infusion pump system **10** can suggest a bolus dosage to the user in a manner that accounts for the user's food intake, the user's recent blood glucose level (e.g., input into the portable pump **60** by the user, from an integral blood test strip analyzer, transmitted to the portable pump **60** from an external blood glucose monitoring device, or the like), the rate of change in the user's blood glucose level, and previously delivered insulin that has not acted on the user. For example, a user can enter a carbohydrate value indicative of a meal into the portable pump **60**, and in response thereto, the portable pump **60** can output a suggested bolus dosage to the user.

[0029] In some embodiments, the infusion pump system **10** may modify a bolus delivery (e.g., a bolus delivery after the user consumes a meal) in response to certain circumstances. For example, the infusion pump system **10** may decrease or otherwise modify a post-meal bolus delivery based on a rapidly falling blood glucose level, a current blood glucose level that is below a threshold

limit, based on an increased level of physical activity, or the like.

[0030] The infusion pump system **10** can be configured to be portable and can be wearable and concealable. For example, a user can conveniently wear the infusion pump system **10** on the user's skin (e.g., using skin adhesive) underneath the user's clothing or carry the portable pump **60** in the user's pocket (or other portable location) while receiving the medicine dispensed from the infusion pump system **10**. As such, the infusion pump system **10** can be used to deliver medicine to the tissues or vasculature of the user in a portable, concealable, and discrete manner.

[0031] Still referring to FIG. **1**, the portable pump **60** includes the user interface **62** that permits a user to monitor and control the operation of the infusion pump system **10**. In some embodiments, the user interface **62** includes a display **65** and the user-selectable buttons (e.g., five buttons **63a**, **63b**, **64a**, **64b**, and **64c** in this embodiment) that are in electrical communication with the control circuitry of the portable pump **60**. For example, the display **65** may be used to communicate a number of status indicators, alarms, settings, and/or menu options for the infusion pump system **10**. In some embodiments, the user may press one or more of the buttons **63a**, **63b**, **64a**, **64b**, and **64c** to shuffle through a number of menus or program screens that show particular status indicators, settings, and/or data (e.g., review data that shows the medicine dispensing rate, the amount of medicine delivered during the last bolus, the delivery time of the last bolus, the total amount of medicine dispensed in a given time period, the amount of medicine scheduled to be dispensed at a particular time or date, the approximate amount of medicine remaining in the cartridge, or the like).

[0032] In some embodiments, the user can adjust the settings or otherwise program the portable pump **60** by pressing one or more buttons **63a**, **63b**, **64a**, **64b**, and **64c** of the user interface **62**. For example, in embodiments of the infusion pump system **10** configured to dispense insulin, the user may press one or more of the buttons **63a**, **63b**, **64a**, **64b**, and **64c** to change the dispensation rate of insulin or to request that a bolus of insulin be dispensed immediately, at a scheduled later time, over a period of time, or following a particular time-based profile. In another example, the user may use the buttons **63a**, **63b**, **64a**, **64b**, and **64c** to manually input information such as the user's current blood glucose level (e.g., as measured by an external blood glucose meter), the current rate of change in the user's blood glucose level, or the like into the portable pump **60**.

[0033] In some embodiments, the NFC circuit **40** is housed in the portable pump **60** to provide an additional functionality that can enhance and simplify user interactions with the portable pump **60**. For instance, using NFC, the need for user activation of multiple buttons **63a**, **63b**, **64a**, **64b**, and **64c** for shuffling through menus may be eliminated or otherwise reduced in some circumstances. In one example depicted in FIG. **1**, the user of infusion pump system **10** has consumed, or will soon consume, a piece of pie that is estimated to include about 60 grams of carbohydrates. As such, the user desires to initiate a corresponding bolus dispensation of insulin to counteract the effects of the intake of 60 grams of carbohydrates. The bolus dispensation of insulin may be intended to cause the user's blood glucose level to remain within a target range.

[0034] To initiate the desired bolus dispensation, the user can first position the portable pump **60** containing the NFC circuit **40** in close proximity with the NFC tag **45** (e.g., preferably within a range 4 inches or less, including for example, a physical “bump” with the NFC tag **45**). Wireless near field communications can thereby be established between the NFC circuit **40** and the NFC tag **45** (as signified by wireless communication symbol **47**). In some embodiments, the user is provided with a notification that near field communications have been established. The notification can be visual, audible, tactile (vibratory), or a combination thereof. In response to the communication between the NFC tag **45** and the portable pump **60**, the portable pump **60** provides a prompt to the user on the display **65**. The prompt on the display **65** requests the user to confirm that the user desires to receive a 4.0-unit dispensation of insulin related to the intake of 60 grams of carbohydrates. To confirm and initiate the dispensation of the suggested bolus amount, the user can simply press button **64c** to select “Enter.” By this example, it can be appreciated that the incorporation of NFC equipment in the infusion pump system **10** can enhance and simplify user

interactions with the infusion pump system **10**, because in order to initiate appropriate suggested bolus dosage of insulin, the user simply bumped the NFC tag **45** with the pump housing structure **66** and then pressed a single acknowledgement button in response to the prompt on the display **65**. As will be described further, in some embodiments, other techniques for user confirmation or acknowledgement can be used, and in some instances user confirmation or acknowledgement may be optional.

[0035] NFC provides short-range wireless communication. As described herein, the maximum working distance for NFC is less than 12 inches, about 8 inches or less, and about 4 inches or less in the aforementioned embodiment depicted in FIG. **1**. NFC allows sharing of relatively small packets of data between a NFC tag and a device equipped with NFC functionality. In the embodiment depicted in FIG. **1**, each NFC tag can store about a kilobyte of data or less, although NFC tags that store a greater quantity of data can also be used in the embodiments described herein. The NFC tags can be configured with a shape that is small and lightweight (e.g., a maximum dimension of about 1 inch or less), particularly because the NFC tags described the embodiment of FIG. **1** do not have an integral power source such as a battery. Instead, a coil in the NFC tag inductively receives magnetic field energy that is emitted from a coil in NFC circuit housed in the portable infusion pump housing structure **66**. Accordingly, energy and data can be wirelessly transmitted between the coils of the NCF tag and the device with NFC functionality. The wireless NFC data transmission can be a two-way wireless communication. That is, data can be transmitted from the NFC tag to the NFC circuit of the portable pump **60**, and data can be transmitted to the NFC tag from the NFC circuit of the portable pump **60**. In other words, the NFC circuit of the portable pump **60** can both read from and write to a NFC tag. The data stored in the NFC tag can be written in a variety of formats. One example format is called the NFC Data Exchange Format (“NDEF”).

[0036] Referring again to FIG. **1**, when the NFC tag **45** communicates with the NFC circuit **40**, the resulting data exchange can trigger one or more automated actions by control circuitry housed in the portable pump **60**. The particular actions are at least in part defined by particular computer program script that is initiated in response to the communications between the NFC tag **45** and NFC circuit **40**. In some arrangements, the particular computer program script is stored on the NFC tag. In such arrangements, when the communications between the NFC tag **45** and NFC circuit **40** are established, the particular computer program script is transferred from the NFC tag **45** to the control circuitry of the portable pump **60** via the NFC circuit **40**. The control circuitry then executes the particular computer program script and can cause the portable pump **60** to automatically perform an action or actions in accordance with the script.

[0037] In alternative arrangements, the particular computer program script to be executed in correspondence to the NFC tag **45** can be stored within the internal control circuitry of the portable pump **60**. In such arrangements, the NFC tag **45** can transfer a unique identifier such as a serial number to the NFC circuit **40**. Upon receipt of the unique identifier, the portable pump **60** can execute the particular computer program script that corresponds to the identifier. In some embodiments, a combination of both arrangements can be used. In any case, from the description herein it can be appreciated that a particular NCF tag (e.g., NFC tag **45**) can be used to automatically trigger a corresponding particular action or change in operation of the portable pump **60**. As such, a variety of NFC tags can be conveniently used with an infusion pump system **10** so as to enhance and simplify user interactions with the infusion pump system **10** in regard to a variety of scenarios and user desires.

[0038] In some embodiments, an accelerometer **50** can be optionally positioned in the portable pump **60** and connected to the control circuitry inside the pump housing structure **66**. In particular embodiments, more than one accelerometer **50** can be included in the pump housing structure **66**. The accelerometer **50** can operate in conjunction with control circuitry and the NFC circuit **40** to supplement the criteria for activating communications between the NFC circuit **40** and the NFC tag

45. In other words, while in some embodiments, communications between the NFC circuit 40 and the NFC tag 45 are activated based merely on the proximity therebetween, in other embodiments a threshold movement of the pump housing structure 66 (as detected by the accelerometer 50) is used (at least as a factor) in activating the NFC circuit 40 for communication with the nearby NFC tag 45. For example, in some embodiments the accelerometer 50 can serve to require that the user “bump” or otherwise tap the portable pump 60 onto the NFC tag 45 or another object before the NFC circuit 40 is activated. An objective for including this feature can be to more clearly ascertain that the user desires to activate NFC when the NFC tag 45 is within the required proximity with the NFC circuit 40. That is, by requiring the user to tap the portable pump 60 onto the NFC tag 45, the user's intentions for activating NFC can be confirmed with a greater level of confidence.

[0039] In some embodiments, this optional feature of using the accelerometer 50 in conjunction with the NFC circuit 40 can function as follows. When a movement is detected by accelerometer 50, the characteristics of the movement can be compared by the control circuitry to a predetermined threshold value (e.g., a threshold movement indicative of the aforementioned “bump” or tap movement). If the detected movement is greater than or equal to the threshold value, the NFC circuit 40 can potentially be activated. But, if no movement that is greater than or equal to the threshold value is detected, the NFC circuit 40 is not activated (even if the NFC circuit 40 is within the required proximity of the NFC tag 45 such that NFC communications can potentially be performed). Therefore, in some embodiments this feature operates to enable NFC when the following two conditions are simultaneously met, or are both met within an establish time interval: (i) an acceleration or an acceleration profile that is greater than or equal to a threshold value is detected (indicating, e.g., a tap or other “bump” action between the portable pump 60 and the NFC tag 45), and (ii) the NFC circuit 40 is in proximity with the NFC tag 45 such that communications therebetween using NFC can occur. In some embodiments, the feature provided by the accelerometer 50 can be activated or deactivated based on a user's or clinician's selection of the feature via the configuration parameters of the portable pump 60. In some embodiments, the accelerometer 50 can be used in conjunction with the NFC circuit 40 in other ways so as to include the detection of a movement into the process for activating or completing changes to the portable pump 60 that correspond to the NFC tag 45.

[0040] In some embodiments, the portable pump 60 can be configured to respond differently when the acceleration threshold value is detected by the accelerometer 50 as compared to when the acceleration threshold value is not detected. For example, as described previously, in response to the detection of the NFC tag 45 by the NFC circuit 40 the user may be asked to confirm via the user interface 62 whether to initiate a change to the portable pump 60, such as initiating a bolus of insulin. However, if an acceleration that meets or exceeds the established threshold is detected by accelerometer 50, and the NFC tag 45 is simultaneously detected (or detected within a threshold time limit) by the NFC circuit 40, in some cases the portable pump 60 may initiate a bolus without requiring additional user confirmation. Still, in some cases additional user confirmation may nevertheless be required before the bolus is initiated.

[0041] Referring now to FIGS. 1 and 2, a set of example NFC tags 46 can be employed for communicating with the portable pump 60 as needed by the user of the infusion pump system 10. For example, some or all of the set of NFC tags 46 can be selected to correspond with the user's commonly performed tasks associated with using the infusion pump system 10. The NFC tags 46 can be distinctly labeled with text, numbers, graphics, colors, textures, braille, photos, symbols, icons, and the like (and combinations thereof) to assist the user to properly and conveniently distinguish between the various types of NFC tags 46. In some embodiments, the NFC tags 46 can have different physical sizes and shapes, and such sizes and shapes can correspond to an amount of carbohydrates associated with the NFC tags 46. In particular embodiments, an assortment of multiple NFC tags 46 can be included on a sheet of flexible plastic film or paper, in a pocket-sized book, on a key ring, in a container, and in many other convenient storage and handling

configurations.

[0042] In one example, a parent may pack a lunch for a diabetic child to take to school, and one or more NFC tags **46** corresponding to the particular lunch food can be packed along with the lunch. Or, the NFC tags **46** can be carried by the child (e.g., in a pocket, worn on a necklace or article of clothing). At the school lunchroom, the child can simply tap the child's infusion pump to the NFC tags **46** in order to command the pump to deliver an appropriate bolus dispensation of insulin in correspondence to the food consumed. Thus, as this example shows, using the NFC tags **46** the user can efficiently, accurately, and conveniently initiate commands to the infusion pump system **10** by activating NFC communications between the infusion pump system **10** and the NFC tags **46** (and, optionally, without the need to input a series of menu selections or other more complex user interface actions). In addition, as will be described further, particular NFC tags **46** can be used to receive and store data from the infusion pump system **10**.

[0043] Some people often eat the same types of foods on a relatively regular basis. A user of the infusion pump system **10** can therefore obtain or make NFC tags that correspond to the food items that the user commonly consumes. For example, the NFC tag **46a** (FIG. 2) for a peanut butter and jelly sandwich could be readily used by a user that regularly consumes such sandwich. As described previously, NFC tag **46a** can have associated with it (either on the NFC tag **46a**, or in the control circuitry of the portable pump **60** in association with a unique identifier of the NFC tag **46a**) data such as the grams of carbohydrates of the food represented by the NFC tag **46a**. In addition to the grams of carbohydrates, the user's preferred way to deliver a corresponding bolus can be included in the data associated with the NFC tag **46a**. For example, the preferred delivery schedule of insulin for the user to counteract the consumption of a peanut butter and jelly sandwich may be 40% of the bolus insulin amount delivered immediately and 60% spread over the next three hours. Of course, a different user may have a different preferred delivery schedule that can be used in correspondence with NFC tags used by the different user. For another type of food item, the preferred delivery schedule of insulin for the user may be other than 40% immediately and 60% spread over the next three hours. For example, for a piece of pie, as represented by the NFC tag **45**, the preferred delivery schedule of insulin for the user may be 50% immediately and 50% spread over the next two hours. As such, the data associated with NFC tag **45** can include the corresponding preferred delivery schedule of insulin of 50% immediately and 50% spread over the next two hours.

[0044] Still further, other data, in addition to grams of carbohydrates and preferred insulin delivery schedules, can be associated with the NFC tags **46**. For example, in some embodiments the fat content, type of fat content, fiber content, protein content, and the like, of the food represented by the NFC tags **46** can be associated with the NFC tags. In some embodiments, such data can be incorporated into a recommended insulin dispensation for the user as calculated by the control circuitry of the portable pump **60**. For example, in some instances meals with increased fat can lead to delayed absorption of the carbohydrates, and thus a bolus determined based on other food contents beyond just carbohydrates, (e.g., fat and protein) may be beneficial.

[0045] While a user of the infusion pump system **10** may consume certain foods like a peanut butter and jelly sandwich fairly regularly, in some circumstances, the user may consume a food item for which the user does not have a dedicated NFC tag **46**. In those circumstances, NFC tags **46b**, **46c**, **46d**, and **46e** can be used if the user so desires. To use the NFC tags **46b**, **46c**, **46d**, and **46e**, the user will estimate the carbohydrate content of the foods that the user has or will soon consume. If, for example, the user will consume food having a carbohydrate content of about 10 grams, the user can activate NFC between the portable pump **60** and the NFC tag **46b** (where the NFC tag **46b** corresponds to 10 grams of carbohydrates). In response, the portable pump **60** may determine a recommended bolus of insulin and either initiate the dispensation of the bolus or prompt the user to confirm via the user interface **62** the initiation of the recommended bolus of insulin. The NFC tags **46c**, **46d**, and **46e** can be similarly used in situations where about 20, 50, or

100 grams of carbohydrates, respectively, have been or will soon be consumed. Of course, the carbohydrate quantities of 10, 20, 50, and 100 grams associated with NFC tags **46b**, **46c**, **46d**, and **46e** are merely illustrative, as NFC tags **46** having any other quantities of carbohydrates (and other data content) can be created and used in accordance with the systems and methods provided herein. [0046] In another example that is relevant to the use of NFC tags **46b**, **46c**, **46d**, and **46e**, it may be determined that the user has or will consume food having a carbohydrate content of about 30 grams. In a first example for handling such a scenario, in some embodiments the portable pump **60** can be configured to add together successive NFC tag data entries to input the total carbohydrate quantity desired by the user. For example, to input 30 grams of carbohydrates, the user may first activate NFC between the portable pump **60** and the NFC tag **46b** to input 10 grams of carbohydrates. Before confirming a bolus dispensation corresponding to the 10 grams, the user can then activate NFC between the portable pump **60** and the NFC tag **46c** to input an additional 20 grams of carbohydrates, for 30 total grams of carbohydrates. In other words, the portable pump **60** can add the first NFC input of 10 grams of carbohydrates and the second NFC input of 20 grams of carbohydrates together to arrive at a total of 30 grams of carbohydrates. The portable pump **60** can then present to the user via the display **65** a prompt that asks the user to confirm the input of 30 grams of carbohydrates to be consumed, and to confirm the acceptance of the associated recommended bolus dispensation of insulin. For example, in the example portable pump **60** provided, the user can confirm the acceptance of such information by activating the button **64c**. In other examples, other techniques for confirming acceptance can be used, as described further herein.

[0047] While the first example immediately above used NFC tags **46b** and **46c** to enter a total of 30 grams of carbohydrates into portable pump **60**, in a second example technique for entering 30 grams of carbohydrates, the single NFC tag **46b** (10 grams of carbohydrates) can be used to activate NFC circuit **40** three times to cause three successive data entries of 10 grams of carbohydrates each. The three successive data entries of 10 grams of carbohydrates each can be added together by portable pump **60** in the manner described above, resulting in a total entry of 30 grams of carbohydrates. The user can then confirm the entry of 30 grams and accept the recommended bolus using the user interface **62**. By way of these examples, it should be appreciated that by combining successive data entries using various NFC tags **46**, such as NFC tags **46b**, **46c**, **46d**, and **46e**, any desired amount of grams of carbohydrates can be entered into portable pump **60** using NFC technology. While in these examples the portable pump **60** was configured to add together successive NFC data entries, in some embodiments the portable pump **60** can alternatively be configured to not add such successive entries together. In some embodiments, the user (or another individual such as a parent or clinician) can selectively configure the portable pump **60** to either add successive entries together or to not add successive entries together.

[0048] Still referring to FIGS. **1** and **2**, NFC tags **46f**, **46g**, **46h**, **46i**, and **46j** are examples of NFC tags that can be conveniently used to enter an estimated quantity of carbohydrates (and optionally other nutritional and operational data) in correspondence to an amount of food consumed, or soon to be consumed, by the user. In general, the NFC tags **46f**, **46g**, **46h**, **46i**, and **46j** can be used as an alternative to counting carbohydrates and entering into the portable pump **60** (via the user interface **62** or via the NFC tags **46b**, **46c**, **46d**, and **46e**) the numerical carbohydrate intake quantity to be consumed (e.g., 10, 20, or 30 grams, etc.). As shown, the NFC tags **46f**, **46g**, **46h**, **46i**, and **46j** can be graduated in relation to an approximate amount of food consumed (e.g., “snack,” “small meal,” “medium meal,” “large meal,” and “extra-large meal”). Such approximations may be appropriate for use by some infusion pump system **10** users or in some situations of using the infusion pump system **10**. Accordingly, when the user presents the NFC tag **46f** (corresponding to a “snack”) to portable pump **60** to activate NFC between the NFC tag **46f** and the portable pump **60**, a lesser quantity of carbohydrates will be input to portable pump **60** in comparison to when the user presents the NFC tag **46i** (“large meal”) to the portable pump **60**. Of course, the NFC tags **46f**, **46g**,

46h, **46i**, and **46j** can be configured to correspond to different levels of carbohydrates for different users. For example, a “large meal” for a male may typically include a greater quantity of carbohydrates than a “large meal” for a female. Therefore, in one example a male user of portable pump **60** may configure (program) NFC tag **46i** to correspond to 200 grams of carbohydrates, while a female user may configure NFC tag **46i** to correspond to 150 grams of carbohydrates. It should be appreciated that the quantity of carbohydrates (and other such data) associated with the NFC tags **46f**, **46g**, **46h**, **46i**, and **46j** can be individualized for the particular user of the infusion pump system **10**.

[0049] NFC tag **46k** is an example of a NFC tag that includes an iconic identifier on a surface of the NFC tag **46k**. In this example, an icon of a hamburger is printed on the NFC tag **46k**. Using icons, symbols, and other types of non-text identifiers can be advantageous for some users. For example, certain users of the NFC tags **46** may not have fluency in the language printed on the NFC tags **46**. Or, a user of the NFC tags **46** may be illiterate, a child, or have poor eyesight. In another example, the NFC tags **46** can include Braille or other raised patterns or shapes for use by blind users or users with limited vision.

[0050] NFC tags **46m** and **46n** are examples of NFC tags that correspond to an exercise activity to be performed by the user of the infusion pump system **10**. Diabetic individuals typically experience a blood sugar reduction in response to the performance of exercise. Therefore, to maintain the user's blood sugar level within a target range it can be beneficial to temporarily reduce the user's basal rate to an extent that correlates to the level of physical exertion performed or to be performed. When reducing basal insulin, the appropriate extent of reduction will depend on factors such as intensity, duration, the individual, and mode of exercise. A basal rate can be reduced prior to, during, and after exercise depending on the situation. For example, in response to performing light exercise over a 30-minute period, the user may present NFC tag **46m** to the user's portable pump **60**. The NFC tag **46m**, for example, may be associated with a command for a 50% reduction of the basal insulin dosages over the next 6 hours. In another example, in response to performing 30 minutes of strenuous exercise, the user may present NFC tag **46n** to the user's portable pump **60**. The NFC tag **46n** may, for example, be associated with a command for a 50% reduction of basal insulin over the next 10 hours. Such factors can be individualized for the particular user, and the particular user's NFC tags **46m** and **46n** can be programmed accordingly. In some embodiments, the NFC tags **46m** and **46n** can be used in combinations to additively arrive at other levels of exertion or duration in a manner analogous to that described above in reference to NFC tags **46b-46e**.

[0051] NFC tags **46o**, **46p**, **46q**, and **46r** are examples of NFC tags that can be used to automate the entry and time-based archival of event occurrences into the portable pump **60**. In other words, the NFC tags **46o-46r** can be used to add descriptive information to the data that is stored within the portable pump **60**. Such labeling of data is also known as data tagging or the creation of metadata. For example, if the user is feeling ill, the user can present the NFC tag **46o** to the user's portable pump **60**. Upon the activation of NFC between the NFC tag **46o** and the NFC circuit **40**, a command is executed that causes the portable pump **60** to store metadata identifying that the user feels ill at the time that the NFC was activated. In other examples, when the user is exercising, eating, or has installed a new medicine cartridge, the user can present the NFC tags **46p**, **46q**, or **46r**, respectively, to the user's portable pump **60**. Upon the activation of NFC between the NFC tags **46p**, **46q**, or **46r** and the NFC circuit **40**, a command is executed that causes the portable pump **60** to store metadata identifying that the user is exercising, eating, or has installed a new medicine cartridge at that time. In another example (not shown in FIG. 2), a NFC tag can be used to indicate when the user has changed the infusion site on the user's body. Accordingly, the presentation of such a NFC tag to the user's portable pump **60** will cause metadata to be stored that identifies that the user changed infusion sites about at the time that NFC was activated between the NFC tag and the NFC circuit **40**.

[0052] NFC tags **46** can also be used to automatically enter other types of commands to the portable pump **60**. NFC tags **46** can thereby reduce the need for using buttons **63a-63b** and **64a-64c** of the user interface **62** to shuffle through various menus. One example of a type of command that can be automated is the entry of a blood glucose reading using a NFC tag **46s**. For example, the user may periodically measure the user's blood glucose level using a blood glucose meter that analyzes a sample of the user's blood using a test strip. The numerical results provided by such a test can then be entered into the user's portable pump **60** to provide the portable pump **60** with the user's actual current blood glucose level. The NFC tag **46s** can be used to “key-up” the portable pump **60** for the entry of the numeric blood glucose level. For example, when the user presents the NFC tag **46s** to the portable pump **60** and NFC is established therebetween, a command is executed that causes the portable pump **60** to get ready to receive the blood glucose data with no other preliminary button pushing required. In such fashion, the user can save time and can operate the infusion pump system **10** with greater convenience using the NFC tag **46s**. Of course, many other types of commands for the portable pump **60** can be similarly automated using the NFC tags **46**.

[0053] NFC tag **46t** is an example of a “blank” NFC tag that can be programmed or scripted and thereafter used to input a variety of commands to the portable pump **60**. In some embodiments, the NFC tag **46t** can be programmed by the portable pump **60**. In particular embodiments, the NFC tag **46t** can be programmed by another device that has NFC functionality (e.g., a smart phone, tablet computer, personal computer, and the like). In some embodiments, the NFC tag **46t** can be written to only once, and thereafter the NFC tag **46t** becomes a read-only NFC tag. In other embodiments, the NFC tag **46t** can be written to, and re-written to, multiple times.

[0054] The programmable NFC tag **46t** can be utilized in a variety of advantageous ways. For instance, as described above a user of the infusion pump system **10** can program the NFC tag **46t** to be associated with data corresponding to a certain type of food that the user consumes (e.g., a large apple having 30 carbs, etc.). In another category of examples, the user can configure the NFC tag **46t** to be used to initiate a particular operation by the portable pump **60**. For example, when changing an infusion set **70** or a medicine cartridge, the user may first want to pause the portable pump **60**. Accordingly, the programmable NFC tag **46t** can be programmed to pause the portable pump **60** if the portable pump **60** is in the run mode at the time that NFC is activated between the programmed NFC tag **46t** and the portable pump **60**. Then, after changing the infusion set **70** or the medicine cartridge, the user may desire to prime the infusion set **70** and begin normal operations of the infusion pump system **70**. Therefore, the programmable NFC tag **46t** can be programmed to prime and thereafter start the portable pump **60** if the portable pump **60** is in the pause mode at the time that NFC is activated between the programmed NFC tag **46t** and the portable pump **60**. In accordance with the examples provided above, it can be appreciated that programmable NFC tag **46t** provides a versatile and customizable functionality whereby users of infusion pump system **10** can enhance and simplify interactions with the user interface **62** and operational capabilities of the portable pump **60**.

[0055] It should be understood from the description herein that a multitude of other beneficial uses for the NFC tags are envisioned for use in combination with a medical infusion pump system. Here, the infusion pump system **70** performs a variety of tasks or receives various types of user entry associated with operating the infusion pump system **70**. Any one of these tasks or types of user entry associated with operating the infusion pump system **70** can be communicated to the control circuitry of the portable pump **60** via the NFC circuit **40** using the corresponding NFC tag. For example, a NFC tag can be used to confirm a user input or pump parameter setting. A NFC tag of this type can be used in conjunction with other NFC tags or input methods to eliminate the need for entering a confirmation using the user interface **62**. In another example, a NFC tag can be used to enter a task command to calibrate a glucose sensor. That is, for example, a NFC tag can trigger the portable pump **60** to use the last blood glucose value entered by the user to calibrate a continuous glucose monitor that is in communication with the infusion pump system **70**. In another

example, in some circumstances, such as when the infusion pump system **70** is used by a child or when the infusion pump system **70** is used during sports activities, it may be desirable to temporarily deactivate the functionality of the buttons **63a**, **63b**, **64a**, **64b**, and **64c** of the user interface **62**. In such circumstances, NFC tags can be used to lock, and subsequently unlock, the buttons **63a**, **63b**, **64a**, **64b**, and **64c** of the user interface **62**. In still another example, a NFC tag can be used to stop or pause the portable pump **60**, such as when the user has disconnected the portable pump **60** from the infusion set **70** to bathe. NFC tags can also be used to enter a task or user command to change to a different basal pattern. Such changes may be beneficial during weekends versus weekdays, during menses versus the rest of the month, and so on. It should be understood that the example uses for NFC tags provided herein are non-limiting, and that other uses for the NFC tags are also envisioned.

[0056] NFC tag **46u** is an example of another use for a “blank” NFC tag that can be written to. In this example, the NFC tag **46u** is used to store the user configuration settings for the user's portable pump **60**. Using the NFC tag **46u** in this manner can provide a way to create a back-up copy of the user's configuration settings. Having a back-up copy of the user's configuration settings can be advantageous in a variety of circumstances. For example, if the user's portable pump **60** is damaged such that a repair is necessitated, the NFC tag **46u** containing the user's settings can be used to conveniently reprogram the repaired portable pump **60** by presenting the NFC tag **46u** to the repaired portable pump **60**. Or, if the user's portable pump **60** is damaged beyond repair, the user's settings can be conveniently uploaded to a replacement portable pump **60** by presenting the NFC tag **46u** to the replacement portable pump **60**. Or, if the user desires different settings for different situations, such NFC tags comprising user settings can conveniently be used to change the settings.

[0057] Referring now to FIG. **3**, the control circuitry of a medical device (e.g., a portable infusion pump in this embodiment) that includes NFC equipment can implement a process **300** of receiving commands from a NFC tag and controlling the medical device in accordance with the commands. Such a process **300**, for example, can be implemented by the control circuitry housed in the portable pump **60** of the infusion pump system **10** (FIG. **1**), and other embodiments of infusion pump systems provided herein (e.g., FIGS. **4**, **5**, **6**, and **7**).

[0058] In operation **310**, the control circuitry of a medical device can receive input via wireless communication from a NFC tag. The input can be indicative of a task associated with using the medical device. A medical device that can perform operation **310** is exemplified in FIG. **1**, where the infusion pump system **10** includes a NFC circuit **40** that is in electrical communication with the control circuitry of the infusion pump system **10**. As explained, the NFC circuit **40** can function to send and receive communications from the NFC tag **45** when NFC is activated by placing the NFC tag **45** within the requisite proximity with the portable pump **60** such that NFC communications are activated.

[0059] In some embodiments, NFC tags can be scripted with executable code that can be transferred to the medical device's control circuitry via the NFC circuit in communication with the control circuitry. In those embodiments, the control circuitry can execute the code as received from the NFC tag. In other embodiments, the NFC tag can communicate a unique identifier, such as a serial number, to the control circuitry via the NFC circuit. In response to the receipt of such a unique identifier by the control circuitry, the control circuitry can execute certain coded operations that are associated with the particular unique identifier received.

[0060] An example of operation **310** is provided in FIG. **1**, where the NFC tag **45** is presented to the NFC circuit **40** of the portable pump **60**. In response, the control circuitry of the portable pump **60** executed commands indicative of an entry by the user of an intent to initiate a bolus dispensation to counteract the consumption of 60 grams of carbohydrates.

[0061] In operation **320**, the control circuitry optionally provides a prompt via the user interface display to confirm a change in operation of the medical device in response to said input from the near field communication tag. Such a prompt may be advantageously used to confirm the user's

intent to change the operation of the medical device before the control circuitry actually implements the change.

[0062] An example of operation **320** is provided in FIG. **1**, where the control circuitry of the portable pump **60** generated the illustrated textual prompt on the display **65**. The prompt provides a description of the potential change in operation (“Consume 60 g of carbs? Press enter to initiate a bolus of 4.0 U.”). By pressing button **64c** the user can confirm the user's intent to implement a change in the operation of the infusion pump system **10**. Alternatively or additionally, other techniques can be used to confirm the user's intent to change the operation of the medical device before the control circuitry actually implements the change. For example, in some embodiments the user can be required to present the same NFC tag to the NFC circuit multiple times within a limited period of time (e.g., two quick taps, three taps within a period of two second, or the like) to confirm the user's intent. In particular embodiments, the user can be required to make contact (e.g., by tapping or otherwise bumping, or the like) between the pump device and the NFC tag, and such contact can be considered to be sufficient user confirmation. In such embodiments, one or more accelerometers in the pump device may be used to detect the requisite contact(s). In other embodiments, some types of tasks entered using a NFC tag require user confirmation while other types of tasks entered using a NFC tag do not require user confirmation. In still other embodiments, a particular task that is entered using a NFC tag will require a user confirmation in some circumstances, but not in other circumstances. An infusion pump system may be configurable to select between the use of these types of alternative techniques for user confirmation. In some embodiments, such various alternatives can be combined for use with various types of tasks associated with a single pump system. In particular embodiments, the infusion pump system can be configured to not require user confirmation for some, or all, tasks and commands entered using NFC tags.

[0063] In operation **330**, after receiving confirmation from the user to implement the change associated with the input from the NFC tag, the control circuitry can control the medical device to change the operation of the medical device in accordance with the user's confirmation of the change. Again, in regard to the example of FIG. **1**, when the user confirms by pressing button **64c** the change to the infusion pump system **10** related to the user's consumption of 60 grams of carbohydrates, the control circuitry can thereafter control the portable pump **60** to deliver the corresponding bolus dispensation of insulin.

[0064] Now referring to FIG. **4**, a schematically represented example portable infusion pump system **400** can include a pump controller device **460** that is equipped with a NFC circuit **463** for providing NFC capabilities. The NFC circuit **463** can be used by the portable infusion pump system **400** to communicate with example NFC tags **410**, **422**, and **432**. Certain items of the portable infusion pump system **400** are shown with dashed lines to indicate that they are optional or alternative items, as explained below.

[0065] The pump controller device **460** includes a control module **461** that can be made up of one or more components. In this embodiment, the control module **461** is configured to communicate control or power signals to the other components of the portable infusion pump system **400**, and to receive inputs and signals therefrom. In some embodiments, the control circuitry can include a main processor board that is in communication with a power supply board. The control circuitry can include at least one processor that coordinates the electrical communication to and from the control module **461** and other components of the portable infusion pump system **400**. For example, the user interface **462** of the pump controller device **460** can include input components and output components that are electrically connected to the control circuitry of the control module **461**. In some embodiments, the control module **461** can receive input commands from a user's button selections (e.g., buttons as shown in FIG. **1**, **5**, **6**, or **7**), and thereby cause the display device of the user interface **462** to output a number of menus or program screens that show particular settings and data (e.g., review data that shows the medicine dispensing rate, the total amount of medicine

dispensed in a given time period, the amount of medicine scheduled to be dispensed at a particular time or date, the approximate amount of medicine remaining the cartridge, the amount of battery life remaining, or the like).

[0066] The processor of the control module **461** can be arranged on a main processor circuit board of the control module **461** along with a number of other electrical components such as computer-readable memory devices. The control circuitry can be programmable in that the user or a clinician may provide one or more instructions to adjust a number of settings for the operation of the portable infusion pump system **400**. Such settings may be stored in the memory devices of the control module **461**. Furthermore, the control module **461** may include one or more dedicated memory devices that store executable software instructions for the processor. The control module **461** may include other components, such as sensors, that are electrically connected to the main processor board. A rechargeable battery pack (not shown) may provide electrical energy to the control module **461**, and to other components of the pump controller device **460** (e.g., user interface **462**, NFC circuit **463**, and others).

[0067] The user interface **462** of the pump controller device **460** permits a user to monitor and control the operation of the pump controller device **460**. For example, the user interface **462** can include a display device having an active area that outputs information to a user, and buttons (e.g., actuatable buttons as shown in FIG. **1**, **5**, **6**, or **7** or touchscreen buttons defined on the display device) that the user can use to provide input. The display device can be used to communicate a number of settings or menu options for the portable infusion pump system **400**. The display may include an active area in which numerals, text, symbols, images, or a combination thereof can be displayed (refer, for example, to FIG. **1**). For example, the user may press one or more buttons to shuffle through a number of menus or program screens that show particular settings and data (e.g., review data that shows the medicine dispensing rate, the total amount of medicine dispensed in a given time period, the amount of medicine scheduled to be dispensed at a particular time or date, the approximate amount of medicine remaining in the cartridge, or the like). In some embodiments, the user can adjust the settings or otherwise program the control module **461** via the user interface **462**. For example, in embodiments of the portable infusion pump system **400** configured to dispense insulin, the user may press one or more of the buttons of the user interface **462** to change the dispensation rate of insulin or to request that a bolus of insulin be dispensed immediately or at a scheduled, later time.

[0068] The pump controller device **460** also includes the NFC circuit **463** in electrical communication with the control module **461**, such that power and signal data can be transferred between the NFC circuit **463** and the control module **461**. The NFC circuit **463** in this embodiment includes a NFC transceiver circuit that is coupled to a loop inductor (e.g., coil) that serves as an antenna for wirelessly communicating with external NFC tags (e.g., NFC tags **410**, **422**, and **432**). The NFC circuit **463** can act as an interface to facilitate the transfer of data between the example NFC tags **410**, **422**, and **432** and the control module **461** using NFC protocols. The coil of the NFC circuit **463** inductively supplies electrical power to the NFC tags **410**, **422**, and **432** by way of secondary coils located in the NFC tags **410**, **422**, and **432** themselves. The respective coils of the NFC circuit **463** and the NFC tags **410**, **422**, and **432** can also wirelessly exchange two-way data transmissions using the same inductive coupling between the coils.

[0069] The example NFC tags **410**, **422**, and **432** depict some additional advantageous ways of configuring NFC tags to be used in conjunction with the portable infusion pump system **400**. In general, NFC tags **410**, **422**, and **432** include a coil and a microchip. The NFC tags **410**, **422**, and **432** act as passive listening devices. But when the NFC circuit **463** is located in the requisite proximity to the NFC tags **410**, **422**, or **432**, the coil of the NFC tags **410**, **422**, or **432** inductively couples with a coil of the NFC circuit **463**. When the coils are inductively coupled, electrical power is supplied to the NFC tags **410**, **422**, or **432** and data can be exchanged between the NFC tags **410**, **422**, or **432** and the NFC circuit **463**. In some embodiments, about a kilobyte of data or more can

be stored in the NFC tags **410**, **422**, and **432** and transferred to the control module **461** via the NFC circuit **463**.

[0070] The NFC tag **410** is an example of a compact and versatile NFC tag. In some embodiments, NFC tag **410** is about the size of a quarter and is flexible. NFC tag **410** can have an adhesive coating on one surface. In some embodiments, the NFC tag **410** can be incorporated into configurations such as pendants, tiles, chips, stickers, and the like.

[0071] The NFC tag **422** is an example of a NFC tag that has been conveniently incorporated onto a keychain **420**. In one example use of such a configuration, the keychain **420** can be attached to a gym bag and the NFC tag **422** can be programmed like the NFC tags **46m** or **46n** of FIG. 2, that indicate that the user is going to perform 30 minutes of light or strenuous exercise respectively. Then, in conjunction with the user's exercise, the user can simply present the NFC tag **422** on the keychain **420** to the user's pump controller device **460** (e.g., tapping the pump controller device **460** against the NFC tag **432** or in proximity to the NFC tag **432**) to enter a command corresponding to the performance of the exercise.

[0072] The NFC tag **432** is an example of a NFC tag that has been conveniently incorporated onto a food package **430**. In this example, the food package **430** is a sandwich container like those commonly available from a vending machine or convenience store. As shown, the NFC tag **432** can be adhered to or otherwise incorporated directly on the exterior of the food package **430**. The NFC tag **432** can be preprogrammed with data corresponding to the contents of the food package **430**. For example, the NFC tag **432** can be programmed with the number of grams of carbohydrates in the sandwich contained in the food package **430**. The user of the portable infusion pump system **400** can conveniently present the NFC tag **432** to the NFC circuit **463** of the pump controller device **460** (e.g., tapping the pump controller device **460** against the NFC tag **432** or in proximity to the NFC tag **432**) to enter a command corresponding to the consumption of the sandwich inside of the food package **430**.

[0073] The pump controller device **460** can also optionally include an accelerometer **464** in electrical communication with the control module **461**. In some embodiments, more than one accelerometer **464** can be optionally included. Embodiments of the pump controller device **460** that include the optional accelerometer **464** can utilize the functionality of the accelerometer **464** in conjunction with the NFC circuit **463**. That is, the accelerometer **464** can operate in conjunction with the control module **461** and the NFC circuit **463** to supplement the criteria for activating or completing communications between the NFC circuit **463** and the NFC tags **410**, **422**, or **432**. In other words, while in some embodiments, communications between the NFC circuit **463** and the NFC tags **410**, **422**, or **432** are activated solely based on the proximity therebetween, in other embodiments a threshold acceleration, as determined by the accelerometer **464**, must also be detected. An objective for including this feature can be to more clearly ascertain that the user intends to activate NFC when a NFC tag **410**, **422**, or **432** is within the required proximity with the NFC circuit **463**. That is, for example, by requiring the user to tap the pump controller device **460** and the NFC tag **410**, **422**, or **432** together, the user's intentions for activating NFC may be ascertained with a greater level of confidence.

[0074] This optional feature using the accelerometer **464** can function as follows. When motion of the pump controller device **460** is detected by accelerometer **464**, a characteristic value of the detected motion can be compared by the control module **461** to a predetermined threshold movement value. If the characteristic value of the detected movement is greater than the threshold value, the NFC circuit **463** can potentially be activated. But, if the characteristic value of the detected movement is not greater than the threshold value, the NFC circuit **463** is not activated (even if the NFC circuit **463** is within the required proximity of the NFC tags **410**, **422**, or **432** such that NFC communications can be performed). Therefore, in some embodiments this feature operates to enable NFC when the following two conditions are simultaneously met, or are both met within an establish time interval: (i) a characteristic value of the detected movement (e.g.,

acceleration value) that is greater than a threshold value is detected (indicating, for example, a tap), and (ii) the NFC circuit **463** is in proximity with the NFC tags **410**, **422**, or **432** such that communications therebetween using NFC can occur.

[0075] Still referring to FIG. **4**, in some embodiments the pump controller device **460** may also serve as the pump unit for the portable infusion pump system **400**, thereby dispensing medicine from the same housing that contains the control module **461** and other components. In those particular embodiments, the pump controller device **460** can be optionally equipped with an integral medicine reservoir and pump drive system **465** in electrical communication with the control module **461**. For example, the portable pump **60** depicted in the embodiment of FIG. **1** is an example of this type of configuration. Such embodiments of the portable infusion pump system **400** can employ a reusable pump apparatus (rather than a disposable pump device as will be described below, for example, in connection with FIG. **5**). Therefore, in those embodiments, the portable infusion pump system **400** may optionally serve as a reusable device that houses the control module **461** and the integral reservoir and pump drive system **465** within a single housing construct. In those circumstances, the pump controller device **460** can be adapted to receive a medicine cartridge in the form of a carpule that is preloaded with insulin or another medicine. The pump drive system **465** can act upon the fluid cartridge to controllably dispense medicine through an infusion set and into the user's tissue or vasculature. In this embodiment, the user can wear the pump controller device **460** on the user's skin under clothing or in the user's pocket while receiving the medicine dispensed through the infusion set. In some embodiments of the pump controller device **460** that include the optional integral reservoir and pump drive system **465**, a refillable medicine reservoir can be incorporated in the pump controller device **460** as an alternative to a medicine cartridge.

[0076] Still referring to FIG. **4**, as an alternative to the integral medicine reservoir and pump drive system **465**, the portable infusion pump system **400** can include a separate pump device **470** (including a reservoir and a drive system) that is in electrical communication with the pump controller device **460**. In these embodiments, the separate pump device **470** can be configured as a disposable and non-reusable pump component while the pump controller device **460** is configured to be reused with a series of the separate pump devices **470**. In the depicted embodiment shown in FIG. **4**, wireless communications are used between the separate pump device **470** and the pump controller device **460**, using a wireless communication module **476** in the pump controller device **460**. The wireless communications of the wireless communication module **476** can utilize any of a variety of wireless communication technologies that have a greater maximum working range than the aforementioned NFC equipment. For example, the wireless communication module **476** can employ BLUETOOTH®, RF (radio frequency), infrared, ultrasonic, electromagnetic induction, and the like, and combinations thereof. Optionally, in some embodiments, the wireless communications of the wireless communication module **476** can utilize NFC equipment. Alternatively, a releasable electrical connection can be used between the separate pump device **470** and the pump controller device **460** so as to provide hardwired electrical communication between the control module **461** of the pump controller device **460** and the drive system of the separate pump device **470**. One such embodiment of a separate pump device **470** that is removably attachable with the pump controller device **460** separate pump device **470** is depicted, for example, in FIG. **5** (as described below).

[0077] In brief, in embodiments of the portable infusion pump system **400** that include the separate pump device **470**, the pump controller device **460** may be configured as a reusable component that provides electronics and a user interface to control the operation of the portable infusion pump system **400**, and the separate pump device **470** can be a disposable component that is discarded after a single use. For example, the separate pump device **470** can be a “one time use” component that is thrown away after the fluid cartridge therein is exhausted. Thereafter, the user can wirelessly connect or removably mount a new separate pump device **470** to the reusable pump controller device **460** for the dispensation of a new supply of medicine from the new separate pump device

470. Accordingly, the user is permitted to reuse the pump controller device **460** (which may include complex or valuable electronics) while disposing of the relatively low-cost separate pump device **470** after each use. Such a portable infusion pump system **400** can provide enhanced user safety as a new separate pump device **470** is employed with each new fluid cartridge.

[0078] Still referring to FIG. **4**, the pump controller device **460** can also optionally include an integral blood strip reader **466** mounted therein and being in electrical communication with the control module **461**. In such embodiments of the pump controller device **460**, test strips (e.g., blood test strips) containing a sample of the user's blood can be inserted into the blood strip reader **466** portion of the pump controller device **460**, to be tested for characteristics of the user's blood. The results of the analysis can be used to affect the dosage or schedule of medicine dispensations from the pump controller device **460** to the user as determined by the control module **461**. As an alternative to or in addition to the internal blood strip reader **466** housed in the pump controller device **460**, the pump controller device **460** can be configured to communicate with an external blood glucose monitor **480**, such as a continuous glucose monitor or a handheld blood glucose meter. For example, the test strips (e.g., glucose test strips) containing a sample of the user's blood can be inserted into external handheld blood glucose monitor **480**, which then analyzes the characteristics of the user's blood and communicates the information (via a wired or wireless connection) to the pump controller device **460**. In other embodiments, the user interface **462** of the pump controller device **460** can be employed by the user to manually enter the user's blood glucose information as reported on a screen of a handheld blood glucose monitor **480**. In still other embodiments, the portable infusion pump system **400** can include a continuous blood glucose monitor **480** (as an alternative to or in addition to the internally housed blood strip reader **466**) that can continuously monitor characteristics of the user's blood and communicate the information (via a wired or wireless connection) to the pump controller device **460**. One example of this configuration is described below in connection with FIG. **6**.

[0079] Referring now to FIG. **5**, some embodiments of an infusion pump system **500** equipped with NFC capabilities can include a removable pump device **100** (shown in an exploded view) and a controller device **200** that communicates with the pump device **100**. The pump device **100** in this embodiment includes a pump housing structure **110** that defines a cavity **116** in which a fluid cartridge **120** can be received. The pump device **100** can also include a cap device **130** to retain the fluid cartridge **120** in the cavity **116** of the pump housing structure **110**. The pump device **100** can include a drive system **140** that advances a plunger **125** in the fluid cartridge **120** so as to dispense fluid **126** therefrom.

[0080] In some embodiments, the controller device **200** communicates with the pump device **100** to control the operation of the drive system **140**. When the controller device **200**, the pump device **100** (including the cap device **130**), and the fluid cartridge **120** are assembled together, the user can (in some embodiments) conveniently wear the infusion pump system **500** on the user's skin under clothing, in a pouch clipped at the waist (e.g., similar to a cell phone pouch), or in the user's pocket while receiving the fluid dispensed from the pump device **100**. Optionally, the controller device **200** may be configured as a reusable component that provides electronics and a user interface to control the operation of the pump device **100**. In such circumstances, the pump device **100** can be a disposable component that is disposed of after a single use. For example, the pump device **100** can be a "one time use" component that is thrown away after the fluid cartridge **120** therein is exhausted. Thereafter, the user can removably attach a new pump device **100** (having a new fluid cartridge **120**) to the reusable controller device **200** for the dispensation of fluid from a new fluid cartridge **120**. Accordingly, the user is permitted to reuse the controller device **200** (which may include complex or valuable electronics, as well as a rechargeable battery) while disposing of the relatively low-cost pump device **100** after each use. Such an infusion pump system **500** can provide enhanced user safety as a new pump device **100** (and drive system therein) is employed with each new fluid cartridge **120**.

[0081] Briefly, in use, the pump device **100** is configured to removably attach to the controller device **200** in a manner that provides a secure fitting, an overall compact size, and a reliable electrical connection that can be resistant to water migration. For example, the controller device **200** can include a controller housing **210** having a number of features that mate with complementary features of the pump housing structure **110**. In such circumstances, the controller device **200** can removably attach with the pump device **100** in a generally side-by-side configuration. The compact size permits the infusion pump system **500** to be discrete and portable. Moreover, at least one of the pump device **100** or the controller device **200** can include a release member that facilitates an easy-to-use detachment and replacement process.

[0082] As shown in FIG. 5, the pump device **100** can include an electrical connector **118** (e.g., having conductive pads, pins, and the like) that is exposed to the controller device **200** and that mates with a complementary electrical connector (not shown) on the adjacent face of the controller device **200**. The electrical connection between the pump device **100** and the controller device **200** provides the electrical communication between the control circuitry housed in the controller device **200** and at least a portion of the drive system **140** or other components of the pump device **100**. For example, in some embodiments, the electrical connection between the pump device **100** and the controller device **200** can permit the transmission of electrical control signals to the pump device **100** and the reception of feedback signals (e.g., sensor signals) from particular components within the pump device **100**. The electrical connection between the pump device **100** and the controller device **200** may similarly facilitate transmission of one or more power signals for sharing electrical power therebetween.

[0083] The pump device **100** may include a drive system **140** that is controlled by the removable controller device **200**. Accordingly, the drive system **140** can accurately and incrementally dispense fluid from the pump device **100** in a controlled manner. The drive system **140** may include a flexible piston rod **141** that is incrementally advanced toward the fluid cartridge **120** so as to dispense the medicine from the pump device **100**. At least a portion of the drive system **140** is mounted, in this embodiment, to the pump housing structure **110**. In some embodiments, the drive system **140** may include a number of components, such as an electrically powered actuator (e.g., reversible motor **142** or the like), a drive wheel **143**, a bearing **145**, the flexible piston rod **141**, and a plunger engagement device **144**. In this embodiment, the reversible motor **142** drives a gear system to cause the rotation of the drive wheel **143** that is coupled with the bearing **145**. The drive wheel **143** may include a central aperture with an internal thread pattern, which mates with an external thread pattern on the flexible piston rod **141**. The interface of the threaded portions of the drive wheel **143** and flexible piston rod **141** may be used to transmit force from the drive wheel to the flexible piston rod **141**. Accordingly, in the embodiment of FIG. 5, the drive wheel **143** is the driver while the flexible piston rod **141** is the driven member. The rotation of the drive wheel **143** can drive the flexible piston rod **141** forward in a linear longitudinal direction to cause the plunger engagement device **144** to nudge the plunger **125** within the fluid cartridge **120** so as to dispense fluid **126** therefrom.

[0084] Still referring to FIG. 5, the controller device **200** can include a user interface **220** that permits a user to monitor and control the operation of the pump device **100**. In some embodiments, the user interface **220** can include a display device **222** and one or more user-selectable buttons (e.g., several buttons **224** are shown in the embodiment of FIG. 5). The display device **222** can include an active area in which numerals, text, symbols, images, or a combination thereof can be displayed. For example, the display device **222** can be used to communicate a number of settings or menu options for the infusion pump system **500**. In this embodiment, the user may press one or more of the buttons **224** to shuffle through a number of menus or program screens that show particular settings and data (e.g., review data that shows the medicine dispensing rate, the total amount of medicine dispensed in a given time period, the amount of medicine scheduled to be dispensed at a particular time or date, the approximate amount of medicine remaining in the fluid

cartridge **120**, or the like). In some embodiments, the user can adjust the settings or otherwise program the controller device **200** by pressing one or more buttons **224** of the user interface **220**. For example, in embodiments of the infusion pump system **500** configured to dispense insulin, the user may press one or more of the buttons **224** to change the dispensation rate of insulin or to request that a bolus of insulin be dispensed immediately or at a scheduled, later time. In some implementations, the display device **222** may also be used to communicate information regarding remaining battery life.

[0085] The controller device **200** can also be equipped with an inspection light device **230**. The inspection light device **230** can provide the user with a tool to illuminate and inspect a targeted location. For example, the inspection light device **230** can be directed at the infusion site on the user's skin to verify that the infusion set is properly embedded, or the inspection light device **230** can be directed at the pump device **100** to illuminate the cavity **116** or other areas. The inspection light device **230** can also be used to notify the user to an alert condition of the infusion pump system **500**. An activation of the inspection light device **230** can thereby provide a visual notification (as an alternative to, or in addition to, the visual notification provided on the display device **222**) to the user that attention to the infusion pump system **500** is warranted.

[0086] The controller device **200** of the infusion pump system **500** also includes a NFC circuit **240**. The NFC circuit **240** can wirelessly communicate with external NFC tags, such as example NFC tags **246a**, **246b**, **246c**, and **246d**. As described previously, such wireless communications using NFC technology can enhance and simplify user interactions with the infusion pump system **500**. For instance, using NFC, the need for user activation of buttons **224** for shuffling through menus may be reduced in some circumstances. FIG. 5 depicts an example scenario to illustrate this principle. In this example scenario, the user of infusion pump system **500** has consumed, or will soon consume, about 50 grams of carbohydrates. As such, the user desires to schedule or initiate a corresponding bolus dispensation of insulin to counteract the effects of the intake of 50 grams of carbohydrates. The bolus dispensation of insulin is intended to cause the user's blood glucose level to remain within a target range. To schedule or initiate the desired bolus dispensation, the user first positions the controller device **200** containing the NFC circuit **240** in close proximity with the NFC tag **246c** (which is programmed to correspond to 50 grams of carbohydrates). Wireless NFC communications can thereby be established between the NFC circuit **240** and the NFC tag **246c** (as signified by wireless communication symbol **247**). In some embodiments, the user is provided with a notification that NFC communications have been established. The notification can be visual, audible, tactile, and a combination thereof. In response to the communication from the NFC tag **246c** to the controller device **200**, the controller device **200** provides a prompt to the user on the display device **222**. In this example, the prompt on the display device **222** requests the user to confirm that the user desires to receive a 3.3 Unit dispensation of insulin because of the intake of 50 grams of carbohydrates. To confirm the dispensation of the suggested bolus amount, the user can simply press the button **224** directly below the word "Enter." By this example, it can be appreciated that the incorporation of NFC technology in the infusion pump system **500** can enhance and simplify user interactions with the infusion pump system **500**, because to initiate an appropriate bolus dosage of insulin the user simply had to present a NFC tag **246c** to the NFC circuit **240** and then press button **224** in response to the prompt on the display device **222**.

[0087] Optionally, the controller device **200** can further include at least one accelerometer **250**. In some embodiments, the accelerometer **250** can be used as a criteria to activate or complete the NFC communications when a characteristic value of a detected movement of the controller device **200** is at or above the threshold level, as previously described above in connection with FIGS. 1 and 4. In such circumstances, the control circuitry housed in the controller device **200** can be configured to determine when the controller housing **210** is "bumped" against one of the NFC tags **246a-246d** so as to activate the NFC transmission via the NFC circuit **240**.

[0088] Referring now to FIG. 6, some embodiments of an infusion pump system **600** configured to

communicate with NFC tags can also be configured to wirelessly communicate with a continuous glucose monitoring device **550**. For example, in this embodiment, the infusion pump system **600** can include an infusion pump assembly **660** used to supply insulin or another medication to a user via, for example, an infusion set **70**. The glucose monitoring device **550** communicates with the infusion pump assembly **660** for the purpose of supplying data indicative of a user's blood glucose level to a control circuitry included in the infusion pump assembly **660**. The infusion pump system **600** can utilize the data indicative of a user's blood glucose level in the calculation of a bolus dosage.

[0089] In this embodiment, the infusion pump assembly **660** includes a housing structure **610** that defines a cavity in which a fluid cartridge **625** can be received. The infusion pump assembly **660** also includes a cap device **630** to retain the fluid cartridge **625** in the cavity of the housing structure **610**. The infusion pump assembly **660** includes a drive system (e.g., described in more detail in connection with FIG. 5) that advances a plunger in the fluid cartridge **625** so as to dispense fluid therefrom. In some embodiments, the dispensed fluid exits the fluid cartridge **625**, passes through a flexible tube **72** of the infusion set **70** to a cannula housing **74**. The dispensed fluid can enter through the skin **20** via a cannula **76** attached to the underside of the cannula housing **74**.

[0090] Still referring to FIG. 6, the glucose monitoring device **550** can include a housing **552**, a wireless communication device **554**, and a sensor shaft **556**. The wireless communication device **554** can be contained within the housing **552** and the sensor shaft **556** can extend outward from the housing **552**. In use, the sensor shaft **556** can penetrate the skin **20** of a user to make measurements indicative of characteristics of the user's blood (e.g., the user's blood glucose level or the like). In response to the measurements made by the sensor shaft **556**, the glucose monitoring device **550** can employ the wireless communication device **554** to transmit data to the control circuitry of the infusion pump assembly **660**.

[0091] In some embodiments, the glucose monitoring device **550** may include a circuit that permits sensor signals (e.g., data from the sensor shaft **556**) to be communicated to the wireless communication device **554**. The wireless communication device **554** can transfer the collected data to the infusion pump assembly **660** (e.g., by wireless communication to a communication device **647** arranged in the infusion pump assembly **660**). In some embodiments, the glucose monitoring device **550** can employ other methods of obtaining information indicative of a user's blood characteristics and transferring that information to the infusion pump assembly **660**. For example, an alternative monitoring device may employ a micropore system in which a laser porator creates tiny holes in the uppermost layer of a user's skin, through which interstitial glucose is measured using a patch. Alternatively, the monitoring device can use iontophoretic methods to non-invasively extract interstitial glucose for measurement. In other examples, the monitoring device can include non-invasive detection systems that employ near IR, ultrasound or spectroscopy, and particular embodiments of glucose-sensing contact lenses. Invasive methods involving optical means of measuring glucose could also be added. In yet another example, the monitoring device can include an optical detection instrument that is inserted through the skin for measuring the user's glucose level.

[0092] Furthermore, it should be understood that in some embodiments, the glucose monitoring device **550** can be in communication with the infusion pump assembly **660** via a wired connection. In other embodiments of the infusion pump system **600**, test strips (e.g., blood test strips) containing a sample of the user's blood can be inserted into a strip reader portion of the infusion pump assembly **660** to be tested for characteristics of the user's blood. Alternatively, the test strips (e.g., glucose test strips) containing a sample of the user's blood can be inserted into a glucose meter device (not shown in FIG. 6), which then analyzes the characteristics of the user's blood and communicates the information (via a wired or wireless connection) to the infusion pump assembly **660**. In still other embodiments, characteristics of the user's blood glucose information can be entered directly into the infusion pump assembly **660** via a user interface **620** on the infusion pump

assembly **660**.

[0093] Still referring to FIG. **6**, the infusion pump assembly **660** includes the user interface **620** that permits a user to monitor the operation of the infusion pump assembly **660**. In some embodiments, the user interface **620** includes a display **622** and one or more user-selectable buttons (e.g., four buttons **624a**, **624b**, **624c**, and **624d** in this embodiment, a different arrangement of buttons in other embodiments, or touchscreen buttons in still other embodiments). The display **622** may include an active area in which numerals, text, symbols, images, or a combination thereof can be displayed. For example, the display **622** may be used to communicate a number of status indicators, alarms, settings, and/or menu options for the infusion pump system **600**. In some embodiments, the display **622** can indicate inform the user of the amount of a suggested bolus dosage, the user's blood glucose level, an indication that the user's blood glucose level is rising or falling, an indication that the bolus dosage suggestion includes a correction for the rate of change in the user's blood glucose level, and the like.

[0094] In some embodiments, the user may press one or more of the buttons **624a**, **624b**, **624c**, and **624d** to shuffle through a number of menus or program screens that show particular status indicators, settings, and/or data (e.g., review data that shows the medicine dispensing rate, the amount of medicine delivered during the last bolus, the delivery time of the last bolus, the total amount of medicine dispensed in a given time period, the amount of medicine scheduled to be dispensed at a particular time or date, the approximate amount of medicine remaining in the fluid cartridge **625**, or the like). In some embodiments, the user can adjust the settings or otherwise program the infusion pump assembly **660** by pressing one or more buttons **624a**, **624b**, **624c**, and **624d** of the user interface **620**. For example, in embodiments of the infusion pump system **600** configured to dispense insulin, the user may press one or more of the buttons **624a**, **624b**, **624c**, and **624d** to change the dispensation rate of insulin or to request that a bolus of insulin be dispensed immediately or at a scheduled, later time. In another example, the user may use the buttons **624a**-**624d** to manually input information such as the user's current blood glucose level (e.g., as measured by an external blood glucose meter), the current rate of change in the user's blood glucose level, or the like into the infusion pump system **600**.

[0095] The infusion pump assembly **660** also includes a NFC circuit **640**. The NFC circuit **640** can wirelessly communicate with external NFC tags, such as example NFC tags **646a**, **646b**, **646c**, and **646d**. As described previously, such wireless communications using NFC technology can enhance and simplify user interactions with the infusion pump system **600**. For instance, using NFC, the need for user activation of buttons **624a**-**624d** for shuffling through menus may be reduced in some circumstances. FIG. **6** depicts an example scenario to illustrate this operational concept. In this example scenario, the user of infusion pump system **600** has consumed, or will soon consume, about 20 grams of carbohydrates. As such, the user desires to initiate a corresponding bolus dispensation of insulin to counteract the effects of the intake of 20 grams of carbohydrates. The bolus dispensation of insulin is intended to cause the user's blood glucose level to remain within a target range. To initiate the desired bolus dispensation, the user first positions the infusion pump assembly **660** containing the NFC circuit **640** in close proximity with the NFC tag **646b** (which is programmed to correspond to 20 grams of carbohydrates). Wireless NFC communications can thereby be established between the NFC circuit **640** and the NFC tag **646b** (as signified by wireless communication symbol **641**). In some embodiments, the user is provided with a notification that NFC communications have been established. The notification can be visual, audible, tactile, and a combination thereof. In some embodiments, in response to the communication from the NFC tag **646b** to the infusion pump assembly **660**, the infusion pump assembly **660** provides a prompt to the user on the display **622**. In this example, the prompt on the display **622** requests the user to confirm that the user desires to receive a 1.3-unit dispensation of insulin because of the intake of 20 grams of carbohydrates. To confirm the dispensation of the suggested bolus amount, the user can simply press the button **624d** directly below the word "Enter." By this example, it can be appreciated that

the incorporation of NFC technology in the infusion pump system **600** can enhance and simplify user interactions with the infusion pump system **600**, because to initiate an appropriate bolus dosage of insulin the user simply had to present a NFC tag **646b** to the NFC circuit **640** and then press button **624d** in response to the prompt on the display **622**.

[0096] Optionally, the infusion pump assembly **660** can further include at least one accelerometer **650**. In some embodiments, the accelerometer **650** can be used to activate or complete the NFC communications when a characteristic value of a detected movement of the infusion pump assembly **660** is at or above the threshold level, as previously described above in connection with FIGS. **1** and **4**. In such circumstances, the control circuitry housed in the infusion pump assembly **660** be configured to determine when the housing structure **610** is “bumped” against one of the NFC tags **646a-646d** so as to activate the NFC transmission via the NFC circuit **640**.

[0097] Referring now to FIG. **7**, some embodiments of an infusion pump system **700** can include an ancillary remote-control device **770** configured to communicate with NFC tags **746** and with an infusion pump assembly **760**. In this example embodiment, the remote-control device **770** is a smartphone. In other embodiments, the remote-control device **770** can be other types of devices such as a tablet computer, laptop computer, a PDA, a custom remote device manufactured specifically for interfacing with the infusion pump assembly **760**, and the like. In this example embodiment, the infusion pump assembly **760** is a single-piece pump unit (similar to the embodiment described above in connection with FIG. **1**). In other embodiments of the infusion pump system **700**, the infusion pump assembly **760** can be configured as a two-piece pump assembly such as the example depicted in FIG. **5**.

[0098] In general, the remote-control device **770** includes a control system for controlling the infusion pump assembly **760**, including user interface components such as touchscreen user interface **775** for allowing a user to receive and provide instructions relative to the infusion pump assembly **760**. The remote-control device **770** also includes a wireless interface **757** for communicating with a wireless interface **752** of the infusion pump assembly **760**. The wireless interfaces **752** and **757** for communicating between the infusion pump assembly **760** and the remote-control device **770** can utilize any of a variety of wireless communication technologies, such as BLUETOOTH®, WI-FI®, RF, infrared, ultrasonic, electromagnetic induction, NFC, or combinations thereof. The infusion pump assembly **760** can be used to dispense insulin or another medication to a user via, for example, an infusion set **70** as described in regard to other infusion pump system embodiments herein.

[0099] In this embodiment, the infusion pump assembly **760** includes a housing structure **710** that defines a cavity in which a fluid cartridge (e.g., an insulin cartridge or other medicine cartridge) can be received. The infusion pump assembly **760** also includes a cap device **730** to retain the fluid cartridge in the cavity of the housing structure **710**. The infusion pump assembly **760** includes a drive system (e.g., described in more detail in connection with FIG. **5**) that advances a plunger in the fluid cartridge so as to dispense fluid therefrom. In some embodiments, the dispensed fluid exits the fluid cartridge, passes through a flexible tube **72** of the infusion set **70** to a cannula housing **74**. The dispensed fluid can enter through the skin **20** via a cannula **76** attached to the underside of the cannula housing **74**.

[0100] In some embodiments, the infusion pump system **700** can be configured to supply scheduled basal dosages of insulin (or another medication) along with user-selected bolus dosages. The basal delivery rate can be selected to maintain a user's blood glucose level in a targeted range during normal activity when the user is not consuming food items. The user-selected bolus deliveries may provide substantially larger amounts of insulin in particular circumstances, such as when the user consumes food items, when the user's blood glucose level increases beyond a safe limit, when the user's blood glucose level rises faster than a threshold rate, or other scenarios in which the blood glucose level requires a significant correction. In some embodiments, the infusion pump system **700** may modify a bolus delivery (e.g., a bolus delivery after the user consumes a meal) in response

to certain circumstances. For example, the infusion pump system **700** may decrease or otherwise modify a post-meal bolus delivery based on a rapidly falling blood glucose level, a current blood glucose level that is below a threshold limit, a detection of a high level of physical activity, or the like.

[0101] Still referring to FIG. 7, in this embodiment, the infusion pump assembly **760** includes the user interface **762** that permits a user to monitor the operation of the infusion pump assembly **760**. In some embodiments, the user interface **762** includes a user interface display **765** and one or more user-selectable buttons (e.g., five buttons **764a**, **764b**, **764c**, **763a**, and **763b** in this embodiment). The user interface display **765** may include an active area in which numerals, text, symbols, images, or a combination thereof can be displayed. For example, the user interface display **765** may be used to communicate a number of status indicators, alarms, settings, and/or menu options for the infusion pump system **700**. In some embodiments, the user interface display **765** can indicate inform the user of the amount of a suggested bolus dosage, the user's blood glucose level, an indication that the user's blood glucose level is rising or falling, an indication that the bolus dosage suggestion includes a correction for the rate of change in the user's blood glucose level, and the like.

[0102] In some embodiments, the user may press one or more of the buttons **764a**, **764b**, **764c**, **763a**, and **763b** to shuffle through a number of menus or program screens that show particular status indicators, settings, and/or data (e.g., review data that shows the medicine dispensing rate, the amount of medicine delivered during the last bolus, the delivery time of the last bolus, the total amount of medicine dispensed in a given time period, the amount of medicine scheduled to be dispensed at a particular time or date, the approximate amount of medicine remaining in the cartridge, or the like). In some embodiments, the user can adjust the settings or otherwise program the infusion pump assembly **760** by pressing one or more buttons **764a**, **764b**, **764c**, **763a**, and **763b** of the user interface **762**. For example, in embodiments of the infusion pump system **700** configured to dispense insulin, the user may press one or more of the buttons **764a**, **764b**, **764c**, **763a**, and **763b** to change the dispensation rate of insulin or to request that a bolus of insulin be dispensed immediately or at a scheduled, later time. In another example, the user may use the buttons **764a**, **764b**, **764c**, **763a**, and **763b** to manually input information such as the user's current blood glucose level (e.g., as measured by an external blood glucose meter), the current rate of change in the user's blood glucose level, or the like into the infusion pump system **700**.

[0103] As an alternative to, or in addition to, using user interface **762** to control the infusion pump assembly **760**, the remote-control device **770** can be used to control the infusion pump assembly **760** in this embodiment. Such an arrangement may be convenient, for example, if the user is wearing the infusion pump assembly **760** in a concealed location under clothing. The remote-control device **770** can wirelessly communicate with the infusion pump assembly **760** via the wireless interfaces **757** and **752**. The wireless communications between the infusion pump assembly **760** and the remote-control device **770** can utilize any of a variety of wireless communication technologies, such as BLUETOOTH®, WI-FI®, RF, infrared, ultrasonic, electromagnetic induction, NFC, and the like, and combinations thereof. Using remote-control device **770**, a user can enter and receive information whereby the user can control the infusion pump assembly **760** using the touchscreen user interface **775** of the remote-control device **770** as an alternative to, or in addition to, using the user interface **762** of the infusion pump assembly **760**. In some alternative embodiments, the infusion pump assembly **760** may be configured without a user interface display **765** or other user interface components for purposes of reducing manufacturing costs, in which case the touchscreen user interface **775** of the remote-control device **770** would serve as the user interface for the infusion pump system **700**.

[0104] The remote-control device **770** also includes a NFC circuit **745**. The NFC circuit **745** can wirelessly communicate with external NFC tags, such as the example NFC tag **746**. As described previously, such wireless communications using NFC technology can enhance and simplify user

interactions with the infusion pump system **700**. For instance, using NFC, the need for user activation of buttons **764a**, **764b**, **764c**, **763a**, and **763b**, or for using the touchscreen user interface **775**, for shuffling through menus may be reduced in some circumstances. FIG. 7 depicts an example scenario to illustrate this principle. In this example scenario, the user of infusion pump system **700** has consumed, or will soon consume, about 60 grams of carbohydrates by eating a piece of pie. As such, the user desires to initiate a corresponding bolus dispensation of insulin to counteract the effects of the intake of 60 grams of carbohydrates. The bolus dispensation of insulin is intended to cause the user's blood glucose level to remain within a target range. To initiate the desired bolus dispensation, the user first positions the remote-control device **770** containing the NFC circuit **745** in close proximity with the NFC tag **746** (which is programmed to correspond to 60 grams of carbohydrates). Wireless NFC communications can thereby be established between the NFC circuit **745** and the NFC tag **746** (as signified by wireless communication symbol **741**). In some embodiments, the user is provided with a notification that NFC communications have been established. The notification can be visual, audible, tactile, and a combination thereof.

[0105] In response to the communication from the NFC tag **746** to the remote-control device **770**, the remote-control device **770** can provide a prompt to the user on the touchscreen user interface **775**. In this example, the prompt on the touchscreen user interface **775** requests the user to confirm whether the user desires to receive a 4.0 Unit dispensation of insulin because of the intake of 60 grams of carbohydrates. To confirm the dispensation of the suggested bolus amount, the user can simply touch the portion of the touchscreen user interface **775** that is labeled "YES." Or, the user can decline the dispensation of the suggested bolus amount by touching the portion of the touchscreen user interface **775** that is labeled "NO." By this example, it can be appreciated that the incorporation of NFC technology in the infusion pump system **700** can enhance and simplify user interactions with the infusion pump system **700**, because to initiate an appropriate bolus dosage of insulin the user simply had to present a NFC tag **746** to the NFC circuit **745** of the remote-control device **770** and then touch the portion of the touchscreen user interface **775** that is labeled "YES."

[0106] Optionally, the remote-control device **770** can further include at least one accelerometer **755**. In some embodiments, the accelerometer **755** can be used to activate the NFC communications when a characteristic value of a detected movement of the remote-control device **770** is at or above the threshold level, as previously described above in connection with FIGS. 1 and 4. In such circumstances, the control circuitry housed in remote-control device **770** be configured to determine when the remote-control device **770** is "bumped" against one of the NFC tags **746** so as to activate the NFC transmission via the NFC circuit **745**.

[0107] While the infusion pump system **700** includes the remote-control device **770** that includes NFC circuit **745**, in some embodiments the infusion pump assembly **760** can also include a NFC circuit **740**. Therefore, the user can alternatively present NFC tags to the infusion pump assembly **760** to input information to the infusion pump assembly **760**. Similarly, the infusion pump assembly **760** can optionally include at least one accelerometer **750** that can be used to activate the NFC communications when a characteristic value of a detected movement of the infusion pump assembly **760** is at or above the threshold level, as previously described above in connection with FIGS. 1 and 4. In such circumstances, the control circuitry housed in infusion pump assembly **760** can be configured to determine when the infusion pump assembly **760** is "bumped" against one of the NFC tags **746** so as to activate the NFC transmission via the NFC circuit **740**.

[0108] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. Accordingly, other embodiments are within the scope of the following claims.

Claims

- 1.** A medical infusion pump system, comprising: a pump assembly that includes a pump device and a controller device, wherein the pump device is operable to receive a fluid cartridge and is in a first housing, the controller device is operable to control the pump device and is in a second housing, and the first housing and the second housing are configured to slidably engage one another; and a mobile device comprising a memory for storing computer-readable instructions to access a dosage calculator application for calculating a suggested medication bolus dosage, the mobile device configured to wirelessly transfer data to the pump assembly to cause the pump assembly to deliver the suggested medication bolus dosage to a user, wherein the mobile device is operable to prompt the user for confirmation of acceptance of a suggested medication bolus dosage before wirelessly transferring data to the pump assembly to cause the pump assembly to deliver the suggested medication bolus dosage to the user.
- 2.** The medical infusion pump system of claim 1, wherein the mobile device is a smartphone.
- 3.** The medical infusion pump system of claim 1, wherein the confirmation of acceptance of the suggested medication bolus dosage by the user is provided by physically bumping the mobile device against the pump assembly.
- 4.** The medical infusion pump system of claim 1, further comprising: an accelerometer wherein movement information from the accelerometer is used by the controller device to verify detection of a bump between the pump assembly and the mobile device.
- 5.** The medical infusion pump system of claim 1, further comprising: a glucose monitoring device configured to communicate to the controller a blood glucose level of the user.
- 6.** The medical infusion pump system of claim 1, wherein the dosage calculator application takes into account previously delivered insulin that has not acted on the user.
- 7.** The medical infusion pump system of claim 1, wherein the dosage calculator application is operable to: provide the user with an option to modify the medication bolus dosage to an amount different from the suggested medication bolus dosage.
- 8.** The medical infusion pump system of claim 1, wherein the pump device is a disposable component of the pump assembly and the controller device is a reusable component.
- 9.** The medical infusion pump system of claim 1, wherein the pump device comprises: one or more anchors that penetrate and retain the fluid cartridge inserted in the pump device to prevent removal of the fluid cartridge.
- 10.** The medical infusion pump system of claim 1, wherein the pump device comprises: a cap device that covers the fluid cartridge, wherein the cap device is non-reversibly attached to the first housing.
- 11.** The medical infusion pump system of claim 1, wherein: the first housing has a number of features that mate with complementary features of the second housing, wherein the second housing of the controller device is operable to removably attach with the pump device in a generally side-by-side configuration.
- 12.** The medical infusion pump system of claim 1, further comprising: a tube extending from the pump assembly to an infusion location.
- 13.** The medical infusion pump system of claim 1, wherein the pump assembly is operable to: provide a confirmation signal of bolus delivery.
- 14.** The medical infusion pump system of claim 1, wherein the pump assembly is operable to: receive data indicative of food consumption prior to determining suggested bolus dosage.
- 15.** A medical infusion pump system, comprising: a pump assembly that includes a pump device, wherein the pump device is operable to receive a fluid cartridge; a controller device, wherein the controller device is operable to control the pump device; and a mobile device comprising a memory for storing computer-readable instructions to access a dosage calculator application for calculating a suggested medication bolus dosage, the mobile device operable to: wirelessly transfer data to the pump assembly to cause the pump assembly to deliver the suggested medication bolus dosage to a

user, and prompt the user for confirmation of acceptance of a suggested medication bolus dosage before wirelessly transferring data to the pump assembly to cause the pump assembly to deliver the suggested medication bolus dosage to the user.

16. The medical infusion pump system of claim 15, wherein the controller device is incorporated in the mobile device.

17. The medical infusion pump system of claim 15, wherein the controller device is incorporated with the pump assembly.

18. The medical infusion pump system of claim 15, further comprising: a glucose monitoring device communicate with the mobile device, wherein the controller device is incorporated in the mobile device.

19. The medical infusion pump system of claim 15, further comprising: a glucose monitoring device communicate with the mobile device, wherein the controller device is incorporated with the pump assembly.
